

Divya Upadhyay

AI Software Engineer | DevOps & Cloud Expert | Data & Automation Specialist

Location: CA | Email: upadhyaydivy10@gmail.com | Phone: +1 (352) 721-1032

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary:

As an **AI Software Engineer** with over **5 years of experience**, I specialize in building intelligent, production-ready systems that blend **AI**, **DevOps**, and **data engineering**. Currently, I'm leading backend and AI initiatives at **Golok Global Inc.**, where I'm driving the development of an **Expedia-style travel platform** and an **LLM-powered chatbot** for a ride-hailing app. My work focuses on delivering scalable backend systems, real-time conversational AI, and infrastructure that accelerates development velocity.

Previously, at **Health Innovators Inc.**, I spearheaded the creation of a **Multi-Agent Telehealth Framework** using **LangChain**, **Vision Transformers**, and **FastAPI microservices**, enabling seamless AI-clinician collaboration. My background also includes automating **CI/CD pipelines** and **ETL workflows** at **Nissan Digital**, and working on data pipelines and cloud migrations at **Trigent Software**.

Backed by a **Master's in Computer & Information Science (University of Florida)** and an industry-proven GitHub portfolio, I bring a blend of engineering rigor and innovation to build impactful AI and automation systems across industries.

Experience:

Software Engineer | Golok Global INC.

June 2025 – Present

- Leading backend development for a scalable **travel platform**, overseeing **end-to-end architecture**, backend services, and API integration using **React**, **Java (Spring Boot)**, and **AWS**.
- Designing and deploying an **AI-powered chatbot** for a ride-hailing service (Uber-like app), leveraging **LLMs**, **LangChain**, and **RAG frameworks** to deliver real-time query resolution and booking assistance.
- Spearheading process optimization by identifying development pain points and delivering tooling, scripts, and **sandbox environments** to streamline onboarding and testing.
- Creating technical documentation and infrastructure blueprints to support agile DevOps teams in accelerating development velocity.
- Implementing secure API gateways, **CI/CD pipelines**, and monitoring dashboards for backend microservices, ensuring low latency and high uptime.

AI Engineer | Health Innovators INC

August 2024 – June 2025

- Designed and deployed **AI-powered conversational agents** using **OpenAI APIs**, leveraging **prompt engineering** strategies to support domain-specific tasks in **telehealth**, including **appointment triage**, **symptom screening**, and **patient query resolution**.
- Spearheading the development of an **AI-enabled telehealth platform**, including a **Multi-Agent Framework** with specialized agents like a **Radiology Agent** to assist clinicians in interpreting reports, suggesting next steps, and automating routine medical queries.
- Led the creation and deployment of **scalable FastAPI microservices**, exposing AI model endpoints with containerization via **Docker** and **Kubernetes**, ensuring high availability and auto-scaling in production.
- Automated the continuous deployment and model refresh cycles using **GitHub Actions**, improving operational efficiency and reducing manual intervention by over **40%**.
- Designed and managed **data pipelines** for chatbot training and evaluation, involving **data preprocessing**, **cleaning**, and **structuring** using **Python (Pandas, NumPy)**; visualized user interaction trends and KPIs via **Tableau dashboards** to identify areas for optimization.
- Transitioned components of the ML stack to **Azure**, with hands-on experience in **Azure Blob Storage**, **Azure Functions**, and **Azure ML Studio**, contributing to the company's **cloud-agnostic strategy**.

- Focused on real-time **user feedback analysis** and **Customer Satisfaction Score (CSAT)** metrics to evaluate model effectiveness, iterating on prompts and flows to improve user experience by over **25%** in A/B testing scenarios.
- Worked closely with designers and product managers to ensure **design-first integration** of AI into user-facing features, enabling seamless collaboration between clinicians and AI systems.

DevOps / Data Engineer | Nissan Digital India LLP, India

September 2018 – June 2022

- Played a key role in the **Next Gen Delivery project**, a flagship initiative aimed at transforming software delivery pipelines across Nissan's digital infrastructure by automating every stage from code commit to production.
- Developed fully automated **CI/CD pipelines** using **Jenkins**, **GitLab CI**, and **GitHub Actions**, cutting release times by **40%** and enabling faster, error-free deployments across multiple microservices.
- Automated environment provisioning and configuration management using **Terraform**, **Ansible**, and **AWS CloudFormation**, supporting zero-touch infrastructure setup for test, staging, and production environments.
- Built robust **monitoring and alerting systems** using **Prometheus**, **Grafana**, and **ELK Stack**, significantly reducing system downtime and response time through proactive performance tracking.
- Designed and maintained **Docker-based containers** and orchestrated them using **Kubernetes**, streamlining application lifecycle management and supporting high availability.
- Created scalable **ETL pipelines** for ingesting and processing vehicle telemetry and business operation data using **Apache Airflow**, **PySpark**, and **AWS Glue**, improving reporting efficiency by 30%.
- **Led a cross-functional** collaboration between developers and QA teams during the **Next Gen Delivery** project, driving efficient alignment across **CI/CD**, **Kubernetes**, and **ETL workflows**, and improving deployment velocity by **25%**
- Partnered with cross-functional teams to build **data quality checks**, ensuring clean, reliable datasets for business intelligence and analytics teams; visualized KPIs via **Tableau dashboards** for leadership reporting.

Data Engineer Intern | Trigent Software, India

January 2018 – September 2018

- Assisted in building and maintaining **ETL pipelines** to process structured and unstructured data using **Python**, **SQL**, and **Apache NiFi**, supporting downstream analytics and reporting use cases.
- Developed and executed **data cleaning** and **preprocessing workflows** using **Pandas** and **NumPy**, ensuring high data quality for machine learning experiments and business dashboards.
- Supported migration of on-premise data systems to **AWS**, gaining hands-on experience with **Amazon S3**, **EC2**, and **RDS**, and contributing to cost-optimized storage strategies.
- Wrote and optimized complex **SQL queries** for data extraction and transformation from relational databases, improving query performance and supporting data validation tasks for senior engineers.
- Assisted in **data visualization and reporting** using **Tableau**, helping the BI team create interactive dashboards for internal stakeholders to analyze KPIs and operational metrics.
- Participated in **data schema design**, collaborating with senior data engineers to define efficient structures for storing and indexing high-volume transactional data.
- Contributed to internal documentation and created **Jupyter Notebooks** demonstrating data workflows and reusable preprocessing logic for use in cross-functional projects.
- Initiated and led weekly peer sessions to share insights on **ETL best practices**, **SQL query optimization**, and introductory concepts in **AWS (S3, EC2, RDS)**, improving intern productivity and technical cohesion.

Projects:

DeepNucleiNet Nuclei Instance Segmentation

March 2024 – April 2024

- Conducted a comparative analysis of instance nuclei segmentation (37,333 nuclei images) by fine-tuning **YOLOv8**, **SAM**, and **Detectron2** frameworks
- Achieved **mAP of 0.49 with Detectron2**, outperforming **SAM (IoU: 0.21)** and **YOLOv8**, validating architectural trade-offs for histopathology image analysis.
- Applied domain-specific fine-tuning and result validation using **Custom metrics** to evaluate precision in edge-case segmentation, contributing to clinical-grade ML research.

Road Accident Analysis

September 2023 – November 2023

- Devised a database schema using **Oracle** to gain insights from Fatality Analysis Reporting System (FARS) datasets.
- Developed an interactive dashboard to handle and process over **1.3 million data points**, leveraging **Angular and Express** for the backend. Implemented advanced SQL queries for optimized data retrieval, delivering real-time insights into fatality causes.

GeoPricePredictor

September 2023 – December 2023

- Designed a **house price prediction ML model** focusing on the integration of intrinsic features and external data sources.
- Established a baseline performance with the **XGBoost model**, achieving an R2 score of **0.7566 on 21** intrinsic features.
- Enhanced predictive accuracy for 21,613 properties across Zoom levels 16, 18, and 19 by integrating Points of Interest (POI) data with image features extracted from **Inception V3** using Transfer Learning. Achieved an **R² score of 0.9** at Zoom level 16.

Gator Taxi Ride Management

March 2023 – April 2023

- Engineered a sophisticated backend for a ride-sharing service using data structures like **Red-Black Trees and Min Heaps**.
- Optimized time complexities for key operations, demonstrating strong algorithmic and development skills. With these enhancements, the system managed a user surge of up to 200%, achieving a 20% reduction in average ride time allocation.

CodeLings: Converting English to Bash Scripts

March 2023 – April 2023

- Fine-tuned a **code15p-220m-bimodal model** for translating **natural language to Bash commands**, implementing a **seq2seq pipeline** with custom dataset processing and achieving a **BLEU score of 0.3939 and precision of 0.55**.
- Deployed on **Hugging Face** to enhance command-line accessibility for users and allow integration for developers.

Skills:	
Programming Languages:	Python, Bash, YAML, JSON, C++, Java, Go, JavaScript, TypeScript
Machine Learning & AI:	Hugging Face Transformers, PyTorch, TensorFlow, Keras, Transfer Learning, Vision-Language Models, RAG, LLMs, Generative AI, BLEU Evaluation, NLP, Model Fine-Tuning, Prompt Engineering, MLflow, Azure ML Studio (Basic), CUDA, Data Parallelism
MLOps & Deployment:	FastAPI, Docker, Kubernetes, GitHub Actions, Jenkins, ML Pipeline Automation, Model Versioning, Model Monitoring, CI/CD, DevSecOps
Cloud Platforms:	AWS (EC2, S3, Lambda, ECS, Redshift, IAM), Azure (Blob Storage, Functions, ML Studio), GCP (BigQuery)
DevOps & Infrastructure:	Terraform, Ansible, Linux, Shell Scripting, Monitoring (Grafana, Prometheus, CloudWatch, ELK), Nexus, SonarQube
Data Engineering:	ETL Pipelines, Pandas, NumPy, Apache Airflow, PySpark, Spark, Kafka, Flink, Cassandra, Redis, MongoDB, PostgreSQL, SQL, NoSQL, Vector Databases (Pinecone, FAISS), Graph Databases (Neo4j)
Tools & Platforms:	Git, GitHub, Jupyter Notebook, VS Code, Postman, Chatbot Frameworks (Rasa – Basic), Design Workflows (Drag/Low-Code), BitBucket, Bamboo, ServiceNow, Tableau, Streamlit
Web & Backend Development:	Django, Flask, Spring Boot, Node.js, Express.js, Next.js, React, Angular

Education:	
•	MS in Computer and Information Sciences University of Florida [May 2024]
•	B-Tech in Computer Science and Engineering KIIT University, India [May 2019]

Certification:

- **System Design Masterclass** | Udemy **[May 2025]**
- **Agentic Design Patterns with AutoGen** | DeepLearning.AI **[October 2024]**
- **Fundamental of Accelerated Data Science** | NVIDIA **[October 2024]**
- **Building Agentic RAG with Llamaindex** | DeepLearning.AI **[September 2024]**
- **Data Parallelism: How to Train Deep Learning Models on Multiple GPUs** | NVIDIA **[September 2024]**
- **Knowledge Graphs for RAG** | DeepLearning.AI **[September 2024]**
- **LangChain Chat with your Data** | DeepLearning.AI **[September 2024]**
- **LangChain for LLM Application Development** | DeepLearning.AI **[September 2024]**
- **Multimodal RAG: Chat with Videos** | DeepLearning.AI **[September 2024]**
- **Fundamentals of Deep Learning** | NVIDIA **[February 2024]**
- **Generative AI with Diffusion Models** | NVIDIA **[February 2024]**
- **Computer Vision Essentials** | Great Learning **[June 2022]**
- **Natural Language Processing: NLP with Transformers in Python** | Udemy **[June 2022]**
- **For Completing 100 Questions on AlgoExpert.io** | AlgoExpert **[March 2022]**
- **Data Warehousing on AWS** | AWS **[December 2021]**
- **CI/CD with Jenkins Ansible Docker Kubernetes** | Udemy **[July 2021]**
- **Container Orchestration using Kubernetes** | Coursera **[February 2021]**
- **Developing a Google SRE Culture** | Coursera **[February 2021]**
- **Monitoring & Telemetry of Production Systems** | Coursera **[February 2021]**
- **Building RESTful APIs with Flask** | LinkedIn **[June 2020]**
- **Learning Django** | | LinkedIn **[June 2020]**
- **Taming Big Data with Apache Spark and Python** | Udemy **[June 2020]**
- **Angular & NodeJS – The MEAN Stack Guide** | | Udemy **[May 2020]**
- **Tableau Desktop III: Advanced** | Tableau **[May 2020]**
- **Tableau Desktop II: Intermediate** | Tableau **[April 2020]**
- **Tableau Desktop I: Fundamental** | Tableau **[April 2020]**
- **Networking and Linux** | RCPL **[May 2017]**